Submission for BCX-ENTRY

OTIS (internal use)

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> **Example** (NIMO Winter 2014/2, 0♣). Determine, with proof, the smallest positive integer c such that for any positive integer n, the decimal representation of the number $c^n + 2014$ has digits all less than 5.

14NIMOW2

Walkthrough. This is quite easy; it's just meant to give a concrete example of a bi-directional problem.

(a) Figure out the answer.

(b) Show that this choice of c actually works.

(c) Manually verify that none of the smaller c work.

> **Example** (HMMT 2016 T4, 0♣). Let $n > 1$ be an odd integer. On an $n \times n$ chessboard the center square and four corners are deleted. We wish to group the remaining $n^2 - 5$ squares into $\frac{1}{2}(n^2 - 5)$ pairs, such that the two squares in each pair intersect at exactly one point (i.e. they are diagonally adjacent, sharing a single corner).
>
> For which odd integers $n > 1$ is this possible?

16HMMTT4

Walkthrough. Let's do some cases first.

(a) Can one do $n = 3$?

(b) Can one do $n = 5$?

(c) Can one do $n = 7$?

In fact, for most n the task is impossible. This is a parity argument: we seek a coloring the cells by black and white (not the usual checkerboard) so that any valid pair has different colors.

(d) Find a coloring of the squares by black and white so that diagonally adjacent squares are opposite colors. (Optionally, find all such colorings.)

(e) Use this to narrow down the set of possible n to two values.

(f) Wrap up the problem using your earlier work.

> **Example** (JMO 2019/1, 0♣). There are $a + b$ bowls arranged in a row, numbered 1 through $a + b$, where a and b are given positive integers. Initially, each of the first a bowls contains an apple, and each of the last b bowls contains a pear. A legal move consists of moving an apple from bowl i to bowl $i + 1$ and a pear from bowl j to bowl $j - 1$, provided that the difference $i - j$ is even. We permit multiple fruits in the same bowl at the same time. The

> goal is to end up with the first b bowls each containing a pear and the last a bowls each containing an apple. Show that this is possible if and only if the product ab is even.

19JMO1

Walkthrough. First we show that if ab is even then the goal is possible. The basic idea is to use induction.

(a) If $\min(a, b) \geq 1$, and a and b are opposite parity, show that in one swap one can reduce from (a, b) to $(a-1, b-1)$.

(b) If $\min(a, b) \geq 2$, and a and b are both even, show that in two swaps one can reduce from (a, b) to $(a-2, b-2)$.

(c) Formulate a set of base cases and complete the proof via induction.

Conversely, we need to show the task is impossible if ab is odd.

(d) Let X denote the number of apples in odd-numbered bowls, and let Y denote the number of pears in odd-numbered bowls. Find a relation between X and Y that does not change under the operation.

(e) Use this to show that the task is impossible when ab is odd.

> **Example** (Shortlist 2012 C1, 0♣)**.** There are n positive integers written in a row. Iteratively, Alice chooses two adjacent numbers x and y such that $x > y$ and x is to the left of y, and replaces the pair (x, y) by either $(y+1, x)$ or $(x-1, x)$. Prove that she can perform at most n^n such steps.

12SLC1

Walkthrough. A weaker result is easier to see:

(a) Look up the term "lexicographic order" if you don't know what it means.

(b) Show the process must terminate, ignoring the bound of n^n.

The hard part is to get the bound n^n, which doesn't depend on how big the numbers are, rather only depends on n itself.

The idea is that we need to pay attention to the *relative order* of the numbers, rather than the numbers themselves. After all, the numbers on the board can be as large as Alice wants.

So for each board state B, we define a permutation π_B on $\{1, \ldots, n\}$ where the number in the ith position of B is the $\pi_B(i)$th smallest number. For example,

$$B = (1337, 42, 2012, 1000, 7) \longmapsto \pi_B = 42531.$$

There is a little wrinkle: what do we do with ties?

(c) Choose a convention for breaking ties, so that each B gives an unambiguous π_B, even if some numbers of B are equal.

(d) Under your convention, is π_B "monotonic" in the lexicographic order?

(e) If you answered "yes" to (d), then prove it. If you answered "no", give a different answer to (c) and try again.

(f) Prove that the process terminates in at most $n!$ steps.

> **Example** (USAMO 2015/4, 0♣). Steve is piling $m \geq 1$ indistinguishable stones on the squares of an $n \times n$ grid. Each square can have an arbitrarily high pile of stones. After he finished piling his stones in some manner, he can then perform *stone moves*, defined as follows. Consider any four grid squares, which are corners of a rectangle, i.e. in positions (i,k), (i,l), (j,k), (j,l) for some $1 \leq i,j,k,l \leq n$, such that $i < j$ and $k < l$. A stone move consists of either removing one stone from each of (i,k) and (j,l) and moving them to (i,l) and (j,k) respectively, or removing one stone from each of (i,l) and (j,k) and moving them to (i,k) and (j,l) respectively.
>
> Two ways of piling the stones are equivalent if they can be obtained from one another by a sequence of stone moves. How many different non-equivalent ways can Steve pile the stones on the grid?

15AMO4

Walkthrough. Despite its placement, this is not an especially quick problem, which is why it is saved as the last walkthrough. It is really rather easy to mess up some details, and moreover, it also takes a while to write-up depending on how you do it.

On the other hand, the statement is pretty long-winded. Here's a tl;dr: stone moves look like the thing below, count the number of equivalence classes under stone moves.

This walkthrough will present the cleanest approach I know of, due to Ankan Bhattacharya. But you should be aware that most solutions to the problem are much worse.

(a) Let r_i denote the number of stones in the ith row, and c_j the number of stones in the jth column. Show that r_i and c_j never change.

(b) What are $\sum r_i$ and $\sum c_j$?

(c) Show that the number of $2n$-tuples of integers $(r_1, \dots, r_n, c_1, \dots, c_n)$ satisfying

$$\sum r_i = \sum c_j = (\text{answer to (b)})$$

is $\binom{n+m-1}{m}^2$.

So the classic mistake is to assume that (c) gives the answer. In truth, this is only the beginning of the problem. To see why, let's agree that the *signature* of a piling is the tuple described in (c).

Thus we have checked that if two pilings are equivalent, then they have the same signature. But this does not mean the number of piling methods is equal to the number of signatures![1]

(d) There are two more statements we have to prove to finish the problem. What are they?

We'll actually now remodel the problem as follows. Forget about the entire grid. Instead, consider a blackboard where we write (x,y) for every stone in row x and column y. Thus there should be exactly m ordered pairs on the blackboard, one for each stone.

Thus, a stone move amounts to switching the y-coordinates of two ordered pairs.

(e) Consider two pilings which have the same signature. Describe an algorithm to reach one from the other, using the blackboard analogy instead of the grid.

(f) Moreover, show that every possible signature is achievable. (Why is this step necessary?)

(g) Put everything together to complete the solution.

[1] For example, here is another invariant: the total number of stones is m, which takes on only one value. But that certainly does not mean the answer is 1.

Practice problems

Instructions: Solve [42♣]. If you have time, solve [55♣].

> It's not denial. I'm just selective about the reality I accept.
>
> Calvin in *Calvin and Hobbes*

Warning: this unit is not short. Tough coach makes the beginners sweat too :P

Problem 1 (NIMO Winter 2014/3, 2♣)

The numbers 1, 2, ..., 10 are written on a board. Every minute, one can select three numbers a, b, c on the board, erase them, and write $\sqrt{a^2+b^2+c^2}$ in their place. This process continues until no more numbers can be erased. What is the largest possible number that can remain on the board at this point?

14NIMOW3

Let $a, b, c, \ldots, j = 1, 2, 3, \ldots, 10$. In the following, we list how the board looks like in each minute.

- First minute: $\sqrt{a^2+b^2+c^2}, d, e, \ldots j$.
- Second minute: $\sqrt{a^2+b^2+c^2+d^2+e^2}, f, g, \ldots j$.
- Third minute: $\sqrt{a^2+b^2+c^2+\cdots+g^2}, h, i, j$.
- Fourth minute: $\sqrt{a^2+b^2+c^2+\cdots+i^2}, j$.

Hence, the biggest number in the fourth minute is clearly the squareroot made with the greatest numbers between $1, 2, \ldots, 10$. Thus, the largest number that can remain at this point is $\sqrt{10^2+9^2+\cdots+2^2} = \sqrt{384}$.

Problem 2 (Shortlist 2018 N1, 5♣)

Determine all pairs (m, n) of positive integers for which there exists a positive integer s such that sm and sn have an equal number of divisors.

18SLN1

Required Problem 3 (USAMO 2011/2, 9♣)

An integer is assigned to each vertex of a regular pentagon so that the sum of the five integers is 2011. A turn of a solitaire game consists of subtracting an integer m (not necessarily positive) from each of the integers at two neighboring vertices and adding $2m$ to the opposite vertex, which is not adjacent to either of the first two vertices. (The amount m and the vertices chosen can vary from turn to turn.) The game is won at a certain vertex if, after some number of turns, that vertex has the number 2011 and the other four vertices have the number 0. Prove that for any choice of the initial integers, there is exactly one vertex at which the game can be won.

11AMO2

Problem 4 (Soviet Union 1978, 3♣)

A token is placed on an $n \times n$ chessboard. Each of two players moves it in turn to an orthogonally neighbouring cell. A player loses if they move the token to a cell that was already visited (this includes the cell the token started with).

(a) If the token starts on a corner cell, for each n, who wins?

(b) If the token starts adjacent to a corner cell, for each n, who wins?

ZC1AD077

Call a and b the two players, where a moves first.

(a)
- n is an odd number. Pair every cell (except the cell where the token was initially at) with exactly one orthogonally neighbouring cell, making 2×1 rectangles. a will always move the token to a rectangle not visited by it yet, and b will always be able to move the token to the other cell of this rectangle. Therefore, b will always be able to make a move, having this way the winning strategy.
- n is an even number. Pair every cell again as described above (without exceptions this time). However, now, the inverse is going to happen, since b will always be the first to move the token to a rectangle not visited by it before. Hence, a will always be able to make a move, having this way the winning strategy.

(b)
- n is an odd number. Pair every cell as described above (except the nearby corner of the initial token location). Thus, we have a similar situation to the one where n is an even number and the token is initially at the corner. We can see that b is never going to be able to go to the corner nearby the initial token's location. Hence, a has the winning strategy.
- n is an even number. Analogous to the situation where the token is initially on a corner cell and n is an even number. Hence, a will always be able to make a move, having this way the winning strategy.

Problem 5 (INMO 2020/3, 3♣)

Consider a nonempty set $S \subseteq \{0, 1, \ldots, 9\}$. It turns out that every sufficiently large positive integer can be written as the sum of two positive integers a and b, where both a and b have decimal digits only in S. How small can $|S|$ be?

20INMO3

Claim 5.1 — $|S| \geq 5$.

Proof. AFTSOF $|S| \leq 4$. Say $|S| = 4$ and $S = \{p, q, r, s\}$ (making it smaller won't affect our argument). As defined by the statement, a and b have their decimal digits in S, including the ones places, that we define as a' and b', respectively. $a' + b' \pmod{10}$ can result, in total, in $\binom{4}{2} + 4$ different numbers: $\{0, 1, 2, \ldots, 9\}$.

Notice that $p + p, q + q, r + r, s + s \pmod{10}$ are all different even numbers, resting only one more even inside $\{0, 1, 2, \ldots, 9\}$: WLOG, say it is $p + q$. Therefore, p, q have the same parity, but p, s and p, r must have different parity, since we don't want more even numbers. Contradiction! In this case, r, s have the same parity, resulting in 6 even numbers inside $\{0, 1, 2, \ldots, 9\}$, even though there are only 5. □

Now, notice the following:

- $0 = 0 + 0$;
- $1 = 1 + 0$;
- $2 = 1 + 1$;
- $3 = 3 + 0$;
- $4 = 3 + 1$;
- $5 = 4 + 1$;
- $6 = 3 + 3$;
- $7 = 4 + 3$;
- $8 = 4 + 4$;
- $9 = 6 + 3$.

Every number from $\{0, 1, 2, \ldots, 9\}$ is a result of the summation of $x, y \in S = \{0, 1, 3, 4, 6\}$. Therefore, every number $n = a + b$, where both a, b have their decimal digits $x, y \in S$. For example: 57241 has the following decimal digits: $\{5, 7, 2, 4, 1\}$, where

$$5 = 4 + 1, 7 = 4 + 3, 2 = 1 + 1, 4 = 3 + 1, 1 = 0 + 1.$$

Hence, $57241 = 44130 + 13111$.

Problem 6 (JMO 2016/4, 3♣)

Find, with proof, the least integer N such that if any 2016 elements are removed from the set $\{1, 2, \ldots, N\}$, one can still find 2016 distinct numbers among the remaining elements with sum N.

16JMO4

The least integer N that satisfies the statement is

$$\sum_{i=2017}^{4032} i = 6049 \cdot 1008 = 6097392.$$

Notice that if we form pairs of numbers from the set with equal sum, there will be at least 3024 such pairs. Even if at most 2016 pairs are destroyed, there will still remain at least 1008 pairs with equal sum, i.e., 2016 numbers. Each pair consists of the xth number from the left and the xth number from the right. This way, each pair has sum $6048 + 1$ as in $(1, 6048), (2, 6047), \ldots, (3024, 3025)$. $6049 \cdot 1008 = 6097392$.

N can't be less than $6,097,392$ because the least possible sum is

$$\sum_{i=1}^{2016} i = 2033136.$$

However, the first 2016 numbers of the set can be removed, changing the least possible sum to

$$\sum_{i=2017}^{4032} i = 6097392.$$

Problem 7 (IMO 1974/4, 5♣)

Consider a sequence $a_1 < a_2 < \cdots < a_p$ of p positive integers. We wish to partition an 8×8 chessboard into p non-overlapping rectangles (whose sides are aligned with that of the grid) such that the ith rectangle has exactly a_i black squares and exactly a_i white squares inside it. Find the largest value of p for which such a decomposition could occur. Moreover, determine all sequences $(a_1, a_2, \ldots, a_p)$ that achieve this maximum.

74IMO4

Problem 8 (USEMO 2021/1, 5♣)

Let n be a positive integer and consider an $n \times n$ grid of real numbers. Determine the greatest possible number of cells c in the grid such that the entry in c is both strictly *greater* than the average of c's column and strictly *less* than the average of c's row.

21USEMO1

Problem 9 (MOP 2011, 5♣)

Seven points A_1, ..., A_7 are marked on a circle γ. We draw all $\binom{7}{2} = 21$ line segments of the form $\overline{A_iA_j}$ for $1 \le i < j \le 7$. They meet at n distinct points in the interior of γ (not on the boundary). How small can n be, over all possible choices of A_i?

11MOPR21

Problem 10 (Tournament of Towns 2005, 3♣)

Originally, every square of 8×8 chessboard contains a rook. One by one, rooks which attack an odd number of other rooks are removed. (Rooks may not jump over other rooks.) Find the maximal number of rooks that can be removed. (A rook attacks another rook if they are on the same row or column and there are no other rooks between them.)

05TTFSA3

Problem 11 (OTIS Mock AIME II 2025/11, by Jiahe Liu, 5♣)

At an informatics competition each student earns a score in $\{0, 1, \ldots, 100\}$ on each of six problems, and their total score is the sum of the six scores (out of 600). Given two students A and B, we write $A \succ B$ if there are at least five problems on which A scored strictly higher than B.

Compute the smallest integer c such that the following statement is true: for every integer $n \ge 2$, given students A_1, ..., A_n satisfying $A_1 \succ A_2 \succ \cdots \succ A_n$, the total score of A_n is always at most c points more than the total score of A_1.

25OIMEII11

Problem 12 (MP4G 2024/14, 3♣)

The set S contains 2024 elements. What is the size of a largest collection C of subsets of S with the property that the intersection of every three subsets in C is nonempty, but the intersection of every four is empty?

24MP4G14

Problem 13 (Slovenia TST 2018, 3♣)

Let $n \geq 2$ be an integer. There are n^2 marbles, each of which is one of n colors, but not necessarily n of each color. Prove that they may be placed in n boxes with n marbles in each box, such that each box contains marbles of at most two colors.

SVNTST

Below we prove the statement is true for nk marbles of at most k colors in k boxes containing n marbles in each.

Clearly the statement works for the base case $k = 1$. By induction over k, suppose the statement works for every k. Let's now prove it works for $k + 1$ either.

Let's divide our solution in two cases:

- There are exactly n marbles of each color. Hence, just fill all the boxes with n marbles of each color, resulting in all the boxes filled with n marbles of a single color in each box.
- Not every color is present in exactly n marbles. Let m be the current number of marbles painted with the less frequently used color and M with the most frequently used one. Hence, we might do the following algorithm:
  1. Fill the first box with m. Clearly $m < n$, meaning that there is yet space to be filled in the box.
  2. Now, take M and fill the first box with $n - m$ marbles, resulting in exactly n marbles in the first box of only 2 colors.
  3. Notice that the first box is done, so it rests now only $(k + 1) - 1 = k$ boxes and colors to work with.

Notice that the total number of marbles now is $n(k + 1) - n = nk$, which satisfies the statement according to the Inductive Hypotheses.

Problem 14 (Kazakhstan 2016, added by Tilek Askerbekov, 3♣)

Let $n \geq 2$ be a positive integer. Prove that all divisors of n can be written as a sequence $d_1, \ldots, d_k$ such that for any $1 \leq i < k$ one of $\frac{d_i}{d_{i+1}}$ and $\frac{d_{i+1}}{d_i}$ is a prime number.

16KAZ101

Problem 15 (Shortlist 2021 A1, 2♣)

Let $n \geq 1$ be an integer, and let A be a subset of $\{0, 1, 2, \ldots, 5^n\}$. If $|A| = 4n + 2$, prove that there exist $a, b, c \in A$ such that $a < b < c$ and $c + 2a > 3b$.

21SLA1

By the sake of contradiction, say there exist a subset $A' = \{m_0, m_1, \ldots, m_{4n+1}\}$ of $\{0, 1, \ldots, 5^n\}$, where $m_0 < m_1 < \cdots < m_{4n+1}$, that doesn't satisfy the statement. Hence,

$$m_{4n+1} + 2m_i \leq 3m_{i+1} \iff \frac{3}{2}(m_{4n+1} - m_{i+1}) \leq m_{4n+1} - m_i$$

when $i = 0, 1, \ldots, 4n + 1$.

Claim 15.1 — $\left(\frac{3}{2}\right)^l (m_{4n+1} - m_{l+i}) \leq m_{4n+1} - m_i$ when $l = 0, 1, \ldots, 4n + i$.

Proof. First, notice that

$$\left(\frac{3}{2}\right)^0 (m_{4n+1} - m_i) \leq m_{4n+1} - m_i.$$

Moreover, by weak induction over l, suppose

$$\left(\frac{3}{2}\right)^{l-1} (m_{4n+1} - m_{i+l-1}) \leq \frac{3}{2}(m_{4n+1} - m_{i+1}) \leq m_{4n+1} - m_i$$

is true. Hence,

$$\left(\frac{3}{2}\right)^{l-1} (m_{4n+1} - m_{l-1+i}) \geq \left(\frac{3}{2}\right)^l (m_{4n+1} - m_{l+i}).$$

Consequently,

$$\left(\frac{3}{2}\right)^l (m_{4n+1} - m_{l+i}) \leq m_{4n+1} - m_i.$$

□

In particular,

$$\left(\frac{3}{2}\right)^{4n-i} (m_{4n+1} - m_{4n}) \leq m_{4n+1} - m_i.$$

Thus, when $i = 0$,

$$\left(\frac{81}{16}\right)^n (m_{4n+1} - m_{4n}) \leq m_{4n+1} - m_0 \leq 5^n$$

Contradiction! $\left(\frac{81}{16}\right)^n > 5^n$ and $m_{4n+1} - m_{4n} \geq 1$.

Required Problem 16 (PAGMO 2021/5, 9♣)

Celeste has an unlimited amount of each type of n types of candy, numbered type 1, type 2, ..., type n. Initially she takes $m > 0$ candy pieces and places them in a row on a table. Then, she chooses one of the following operations (if available) and executes it:

- She eats a candy of type k, and in its position in the row she places one candy type $k-1$ followed by one candy type $k+1$, indices modulo n.
- She chooses two consecutive candies which are the same type, and eats them.

Find all positive integers n for which Celeste can leave the table empty for any value of m and any configuration of candies on the table.

21PAGMO5

Problem 17 (IMO 1997/4, 3♣)

An $n \times n$ matrix whose entries come from the set $S = \{1, 2, \ldots, 2n-1\}$ is called a *silver* matrix if, for each $i = 1, 2, \ldots, n$, the i-th row and the i-th column together contain all elements of S. Show that:

(a) there is no silver matrix for $n = 1997$;

(b) silver matrices exist for infinitely many values of n.

97IMO4

Problem 18 (JMO 2010/1, 2♣)

Let $P(n)$ be the number of permutations $(a_1, \ldots, a_n)$ of the numbers $(1, 2, \ldots, n)$ for which ka_k is a perfect square for all $1 \le k \le n$. Find with proof the smallest n such that $P(n)$ is a multiple of 2010.

10JMO1

Problem 19 (USAMTS 5/3/29, 5♣)

David has a deck of $2n$ distinct cards, each with one of n colors so that each color appears twice. He deals the cards face-up on a table, one at a time. If there are ever two cards of the same color both face-up at once, he removes both of them.

Of the $(2n)!$ possible original orderings of the deck, determine how many have the following property: at every point, the table contains cards of at most two distinct colors.

USMT5329

Required Problem 20 (Canada 2019/3, 5♣)

Let m and n be positive integers. A $2m \times 2n$ grid of squares is colored in the usual chessboard fashion. Determine the number of ways to place mn counters on the *white* squares, at most one counter per square, so that no two counters are diagonally adjacent.

19CAN3

Problem 21 (EGMO 2018/4, 5♣)

Let $n \ge 3$ be an integer. Several non-overlapping dominoes are placed on an $n \times n$ board. The *value* of a row or column is the number of dominoes that cover at least one cell of that row or column. A domino configuration is called *balanced* if there exists some $k \ge 1$ such that every row and column has value k.

Prove that a balanced configuration exists for every $n \ge 3$ and find the minimum number of dominoes needed in such a configuration.

18EGMO4

Required Problem 22 (JMO 2021/4, 5♣)

Carina has three pins, labeled A, B, and C, respectively, located at the origin of the coordinate plane. In a *move*, Carina may move a pin to an adjacent lattice point at distance 1 away. What is the least number of moves that Carina can make in order for triangle ABC to have area 2021?

21JMO4

Problem 23 (USAMO 2023/5, 3♣)

Let $n \geq 3$ be an integer. We say that an arrangement of the numbers 1, 2, ..., n^2 in an $n \times n$ table is *row-valid* if the numbers in each row can be permuted to form an arithmetic progression, and *column-valid* if the numbers in each column can be permuted to form an arithmetic progression.

For what values of n is it possible to transform any row-valid arrangement into a column-valid arrangement by permuting the numbers in each row?

23AMO5

Required Problem 24 (USAMO 2008/4, 5♣)

For which integers $n \geq 3$ can one find a triangulation of regular n-gon consisting only of isosceles triangles?

08AMO4

(Any set of $n-3$ diagonals of $\mathcal{P}$ that do not intersect in the interior of the polygon determine a *triangulation* of $\mathcal{P}$ into $n-2$ triangles.)

Problem 25 (XOOK 2025/1, 9♣)

For the Exeter disco party, Oron wants to tile a 2024×2024 square in the middle of the dance floor with 1×1 tiles which have arrows pointing either up, left, right, or down. After placing down all the tiles in some configuration, he realizes that he wants to rotate some of the tiles so that no matter which tile a dancer starts on, after following the arrows they end up leaving the grid through some fixed cell. What's the minimum number of tiles he needs to rotate so that this is always possible?

25XOOK1

Problem 26

Let x, y, a, b, c be integers.

1. Prove that $2x + 3y$ is divisible by 17 if and only if $9x + 5y$ is divisible by 17.
2. If $4a + 5b - 3c$ is divisible by 19, prove that $6a - 2b + 5c$ is also divisible by 19.

1. Say
$$17 \mid 2x + 3y \iff 17 \mid k(2x + 3y).$$
However, $k(2x + 3y) - 9x - 5y = x(2k - 9) + y(3k - 5)$. Notice that $k = 13$ results in $17x + 34y$, which is a multiple of 17. Thus, if $2x + 3y$ is divisible by 17, $9x + 5y$ is either. An analogous solution proves the if and only if.
2. Analogous argument proves the statement.

Problem 27

Define the nth Fermat number F_n by $F_n = 2^{2^n} + 1$. Show that F_m, F_n are coprime for any m, n

Claim 27.1 — $(2^{2^m} + 1) \equiv 2 \pmod{(2^{2^n} + 1)} \quad \forall m > n.$

Proof. Let $a = 2^{2^n}$. Hence, we must prove

$$a + 1 \mid 2^{2^m} - 1.$$

Notice $2^{2^m} = (2^{2^n})^{2^{m-n}} = a^{2^{m-n}}$. Since $m < n$, say $a^{2^{m-n}} = 2t$. Therefore, it suffices to prove that

$$a + 1 \mid a^{2^{m-n}} - 1 = a^{2t} - 1 = (a+1)(a^{2t-1} + a^{2t-2} - \cdots + 1) \qquad \square$$

Therefore, F_m and F_n are clearly coprime.